

Minimally Model-Based Trajectory Tracking and Variable Impedance Control of Flexible-Joint Robots

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Abstract—This article presents a novel control scheme that primarily provides a solution to the problem of simultaneously performing stable trajectory tracking and variable impedance control (VIC) of flexible-joint robots. The existing asymptotically stabilizing tracking controllers for compliant robots rely upon several model-based terms and their derivatives. Contrarily, the proposed minimally model-based trajectory tracking control scheme requires knowledge of only the gravitational torque vector model. Although interaction and tracking performance enhancements are achievable through VIC, its operation introduces energy injections, thereby impacting closed-loop stability/passivity. Existing VIC passivity preservation techniques deal exclusively with rigid-joint robots. To this end, the proposed method introduces the minimally model-based trajectory tracking variable impedance control scheme, which ensures passivity/stability when applying VIC to flexible-joint robots. Numerical and experimental results corroborate the controllers' efficacy in terms of realizing VIC interaction and tracking tasks, thus enabling flexible-joint robots to stably replicate biologically inspired behaviors.

Index Terms—Flexible-joint robots, human–robot interaction, impedance control.

I. INTRODUCTION

THE increasing demand for robotic devices within human-inhabited environments has stimulated numerous research efforts aimed to resolve the human–robot interaction problem. The existing methodologies can generally be divided into two distinct categories, namely those of an algorithmic nature that relate to software, and those pertaining to hardware design.

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In terms of software, robotic interaction predominantly entails usage of impedance/admittance control schemes [1], which obviate the need for modeling the entire environment, as would be the case for direct force control techniques. Impedance control was introduced to the field by Hogan [2] as a means of allowing robots to establish user-defined dynamical relationships between their motions and the associated external forces [3]. Robotics literature contains an array of works involving impedance control, including some of the pioneering exertions [2], [3], as well as additional examples of robustly performing impedance controllers [4], [5].

It has been posited that further interactional augmentation can be achieved by real-time modulation of a system's active impedance gains (VIC, variable impedance control) [6]. In fact, Buchli *et al.* [6] suggest that biological machines display superior performance in comparison to their artificial counterparts as a result of their inherent VIC mechanisms. Ikeura and Inooka [7] describe a method by which a human's musculoskeletal system may be represented via impedance parameters, which are modulated in real-time (VIC) when executing a cooperative task with a robot. Tsumugiwa *et al.* [8] propose a viscosity modulation strategy to complement a variable active stiffness controller to ensure successful execution of a human–robot cooperative calligraphic task. A contact-agnostic algorithm for learning impedance parameter adaptation is delineated in [9], while [10] contrives a means of generating Cartesian impedance gains via linear quadratic regulator (LQR) methods. Li *et al.* [11] report on an adaptive impedance control technique whose gain modulation strategy relies on solving the LQR problem via iterative learning. Similarly, Matinfar and Hashtrudi-Zaad [12] describe an impedance adaptation methodology that functions by regulating a controller's active compliance gains as per the environmental parameters. Gribovskaya *et al.* [13] utilize an impedance regulation technique to enhance a learnt task model.

Robotic hardware design research aims to ameliorate a robot's interactional performance via incorporation of elastic elements into its physical structure [14], [15]. If these flexible elements are embedded into joints connecting rigid links, then the resulting structure is referred to as a flexible-joint robot [14], while if a device's links are flexible per se, then it belongs to the category of flexible-link robots [15].

The former category includes robots powered by series elastic actuators (SEAs) [16], [17], whose inherent flexibility renders robotic systems compliant during interaction. However, the

usage of SEAs precludes real-time passive stiffness regulation, although this has been addressed through development of variable stiffness actuators (VSAs) [14], [18]. VSAs could further improve interactional performance, however at the expense of increased design cost, size, and control complexity. Nevertheless, since these designs entail incorporation of passive elastic elements, this also leads to underactuation, which exacerbates the position control problem [19]. Robotic devices powered by harmonic drives also belong to the category of flexible-joint robots [19]–[23].

Other works focus on amalgamating structural and software-induced compliance via implementation of impedance controllers on flexible-joint robots for the purpose of augmenting interactional performance [19], [24]. However, several works introducing stabilizing control schemes for flexible-joint robots [19], [20], [25], [26] augmented the state-of-the-art via incorporation of motor inertia scaling terms, torque, and torque derivative feedback in addition to passivity analyses. Since impedance controllers are customarily used for interaction tasks, their interaction capabilities are frequently assessed via passivity analyses [21], [27].

In terms of globally asymptotically stable (GAS) tracking schemes for flexible-joint robots, the associated literature is somewhat limited [23], [28]–[31], while in most cases the stability results pertain solely to local stability. The works that do offer global asymptotical stability (GLAS) results [22], [23], [29], however, depend upon exceedingly model-based solutions. Furthermore, all the aforementioned works deal with linear stiffness robots [29], with the exceptions of [29], [32]. Contrarily, the minimally model-based tracking control (MMTC) scheme achieves GLAS by using only the gravitational torque vector (without requiring models of inertial, Coriolis, damping, and stiffness terms), and is also applicable to robots with nonlinear passive stiffnesses.

The implementation of active VIC schemes on SEA-powered robots has been reported in other treatises [33]–[35]. Active VIC of compliant robots powered by hydraulic actuators is described in [36], [37], although both works omit stability analyses. Moreover, Hussein *et al.* [38] propose an adaptive impedance controller for a dynamic rehabilitation exoskeleton, which is powered by pneumatic actuators. Al-Shuka *et al.* [39] synopsize the existing active impedance control techniques, thereby revealing that provably stable VIC of flexible-joint robots (the topic of the work presented herein) remains an open problem.

However, VIC of flexible-joint robots is also attainable “mechanically,” via VSAs as demonstrated on DLR’s David [14], and analyzed in [24].

Despite the benefits of VIC [40], this technique may jeopardize a system’s stability/passivity as it tends to induce energy-injecting terms [41]. To this end, Ferraguti *et al.* [41] propose a tank-based approach to suppress the relevant energy injections, and thus preserve passivity. Ferraguti *et al.* [42] offer an extension to [41] by considering a time-varying admittance controller, whose passivity is analyzed using port-Hamiltonians. Contrarily, Kronander and Billard [40] attempt to resolve the energy-injection problem through a modified Lyapunov function whose validity necessitates constraining the active impedance gain fluctuations. However, the aforesaid methods are

inapplicable to flexible-joint robots in contrast to MMTC/MMTVIC, which is applicable to both robot categories. The MMTVIC scheme eliminates energy injections through a power-shaping signal (PSS), without employing energy tanks [41], [42], and obviates the need for constraining the rate of impedance gain modulation [40].

The presented work deals with passivity/stability preservation during VIC, assuming the existence of an arbitrary impedance trajectory. Thus, the MMTVIC scheme enables usage of gain-scheduling methods [43] on flexible-joint robots, in addition to making the following contributions:

- 1) Asymptotically stable tracking control of flexible-joint robots relying only on the gravitational torque model. The existing asymptotically stabilizing, compliant robot tracking controllers rely heavily on model-based terms [22], [23], and high-order position derivatives rendering their performance susceptible to modeling uncertainties.
- 2) A stable active VIC scheme for trajectory-tracking control of flexible-joint robots. There exists one analogous VIC method, which however only applies to regulation [44].
- 3) Direct usage of intuitive full-state feedback, comprising motor- and link-side, position, and velocity gains. Albuschäffer *et al.* [21] deal exclusively with the regulation problem, and use full-state gains in the form of torque and torque derivative feedback. The methods proposed herein enable minimization of link-side errors, while allowing for stable trajectory-tracking control.
- 4) Stable VIC involving real-time modulation of both collocated and noncollocated feedback gains.
- 5) Applicability to robots with variable active and passive stiffnesses, while ensuring stable trajectory tracking. There exist two other methods that deal with VIC involving solely passive stiffness fluctuations [29], [32]. Moreover, Spyraikos-Papastavridis *et al.* [44] only address the problem of regulation.
- 6) The PSS approach may serve as a general solution to underactuated robot control [45]. Rudra *et al.* [46] posit that a block backstepping approach could potentially serve as a general solution to underactuation, however this method is heavily model-based and might be incapable of ensuring GLAS for underactuated systems with contacts.

The remainder of this article is structured as follows. Section II introduces the flexible-joint model and problem statement, while Section III provides the MMTVIC scheme’s stability/passivity analyses. Furthermore, Section IV describes the full-state feedback controllers, Section V presents simulation and experimental results procured from virtual and real collaborative robots, and Section VI concludes this article.

II. COMPLIANT ROBOT IMPEDANCE CONTROL

A. Flexible-Joint Robot Dynamics

An n -degree-of-freedom (DoF), flexible-joint robot, may be modeled via the approach described in [28], yielding

$$\mathbf{M}(\mathbf{q}) \ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{N} \dot{\mathbf{q}} + \mathbf{K}(\mathbf{q} - \boldsymbol{\theta}) + \boldsymbol{\tau}_{gq} = \boldsymbol{\tau}_{ex} \quad (1)$$

$$\mathbf{J} \ddot{\boldsymbol{\theta}} + \mathbf{D} \dot{\boldsymbol{\theta}} - \mathbf{K}(\mathbf{q} - \boldsymbol{\theta}) = \boldsymbol{\tau}_m \quad (2)$$

where $\mathbf{q} \in \mathbb{R}^n$ and $\boldsymbol{\theta} \in \mathbb{R}^n$ denote the link and motor positions, $\mathbf{M}(\mathbf{q}) \in \mathbb{R}^{n \times n}$ is a symmetric, positive definite link inertia matrix, while $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ and $\mathbf{N} \in \mathbb{R}^{n \times n}$ represent the Coriolis/centripetal, and diagonal viscous damping matrices, respectively. $\boldsymbol{\tau}_{gq} = \boldsymbol{\tau}_g(\mathbf{q})$, $\boldsymbol{\tau}_{ex}$, and $\boldsymbol{\tau}_m \in \mathbb{R}^n$ signify the gravitational, external, and actuation torque vectors, respectively. Moreover, $\mathbf{K} \in \mathbb{R}^{n \times n}$ represents a symmetric stiffness matrix, while $\mathbf{J}, \mathbf{D} \in \mathbb{R}^{n \times n}$ signify the symmetric, positive definite motor inertia and damping matrices.

B. Tracking and Impedance Control of Compliant Robots

The majority of compliant robot control schemes entail utilization of PD-plus-type controllers that rely predominantly upon model-based terms, although to varying degrees. Typically, tracking controller designs involve several model-based terms (inertial, damping, Coriolis, and gravitational elements) in conjunction with their derivatives, which are accompanied by high-order position derivatives (jerk and snap).

In terms of interaction control, Ott *et al.* [19] and Albu-Schäffer *et al.* [21] describe how utilization of inner-torque loops in combination with superimposed PD-type control schemes permits construction of high-fidelity impedance controllers. Archetypal impedance control schemes [19], [25] for compliant-joint robots have customarily focused on resolving the regulation problem [19], [25], [26] (among many others), while a limited number of works also address trajectory tracking [20], [22], [23], [29], [30].

C. Problem Statement

Although impedance controllers offer seamless execution of interaction tasks [9], [19], manifold interactional benefits are attainable through VIC [6], [40]. However, stability preservation during active tracking VIC of robots powered by SEAs (or harmonic drives) remains unaddressed.

By utilizing the Lyapunov function and control law presented in [25], one acquires the following Lyapunov derivative expression, when considering (1) and (2):

$$\dot{V}(\dot{\mathbf{q}}_f) = -\dot{\mathbf{q}}_f^T \boldsymbol{\Psi}_P \dot{\mathbf{q}}_f \quad (3)$$

where $\boldsymbol{\Psi}_P \in \mathbb{R}^{2n \times 2n} = \text{diag}(0, \mathbf{D} + \mathbf{K}_{D_M})$, $\dot{\mathbf{q}}_f = [\dot{\mathbf{q}}^T \dot{\boldsymbol{\theta}}^T]^T$, with $\mathbf{K}_{D_M} \in \mathbb{R}^{n \times n}$ denoting a positive-definite damping gain matrix. However, the performance of VIC and, in particular, joint stiffness modulation yields the result

$$\dot{V}(\dot{\mathbf{q}}_f, \boldsymbol{\theta}_E) = -\dot{\mathbf{q}}_f^T \boldsymbol{\Psi}_P \dot{\mathbf{q}}_f + \frac{1}{2} \boldsymbol{\theta}_E^T \dot{\mathbf{K}}_{P_M} \boldsymbol{\theta}_E \quad (4)$$

where $\mathbf{K}_{P_M} \in \mathbb{R}^{n \times n}$ signifies a time-varying active stiffness gain matrix, and $\boldsymbol{\theta}_E = \boldsymbol{\theta}_d - \boldsymbol{\theta}$. The indeterminate nature of (4)'s sign necessitates elimination of the energy-injecting term. The challenge is thus threefold as it entails development of a unified tracking/interaction control scheme [29], which simultaneously preserves closed-loop stability/passivity during VIC while possessing a limited reliance upon model-based terms and high-order matrix and position derivatives (i.e., acceleration, jerk, and snap [29], [30]).

III. MINIMALLY MODEL-BASED TRACKING CONTROL

A. Gravitational-Torque-Only Tracking Controller

To realize stable tracking of time-varying trajectories, when utilizing flexible-joint robots possessing underactuated DoFs, the following joint-impedance MMTC law is proposed:

$$\boldsymbol{\tau}_m = \mathbf{K}_{P_M} \boldsymbol{\theta}_E - \mathbf{K}_{D_M} \dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\theta}}_S^T \mathbf{I} (\dot{\mathbf{q}}^T \boldsymbol{\tau}_{gq} - \dot{\boldsymbol{\theta}}_d^T \mathbf{K}_{P_M} \boldsymbol{\theta}_E) \quad (5)$$

where $\dot{\boldsymbol{\theta}}_S \in \mathbb{R}^{n \times n} = \text{diag}(\dot{\boldsymbol{\theta}})^+$, $\mathbf{I} = [1 \dots 1_n]^T \cdot \text{rank}(\dot{\boldsymbol{\theta}}_S)^+$, with "+" denoting the Moore–Penrose pseudo-inverse. $\mathbf{K}_{P_M}$ and $\mathbf{K}_{D_M} \in \mathbb{R}^{n \times n}$ denote constant, diagonal, positive definite motor-side PD gain matrices. The PSS that differentiates this controller from other PD-type schemes is perceived as an additional damping element. To prove closed-loop stability during trajectory tracking, assuming $\boldsymbol{\theta}_d(t)$ is continuously differentiable the following Lyapunov function is utilized:

$$V = \frac{1}{2} \dot{\mathbf{q}}_f^T \mathbf{M}_T \dot{\mathbf{q}}_f + \frac{1}{2} \mathbf{q}_f^T \mathbf{K}_T \mathbf{q}_f + \frac{1}{2} \boldsymbol{\theta}_E^T \mathbf{K}_{P_M} \boldsymbol{\theta}_E \quad (6)$$

where $\mathbf{M}_T \in \mathbb{R}^{2n \times 2n} = \text{diag}(\mathbf{M}, \mathbf{J})$ and $\mathbf{K}_T \in \mathbb{R}^{2n \times 2n} = \begin{bmatrix} \mathbf{K} & -\mathbf{K} \\ -\mathbf{K} & \mathbf{K} \end{bmatrix}$ denote the inertia and passive stiffness matrices, respectively.

Theorem 1: The closed-loop system comprising (1), (2), (5), is GAS when supplied with trajectories defined by the $\boldsymbol{\theta}_d(t)$ variable (tracking control), while assuming that $\boldsymbol{\tau}_{ex} = 0$.

Proof 1: Calculating the time derivative of (6), initially produces the expression

$$\dot{V} = \frac{\dot{\mathbf{q}}_f^T \dot{\mathbf{M}}_T \dot{\mathbf{q}}_f}{2} + \dot{\mathbf{q}}_f^T \mathbf{M}_T \ddot{\mathbf{q}}_f + \dot{\mathbf{q}}_f^T \mathbf{K}_T \dot{\mathbf{q}}_f + \dot{\boldsymbol{\theta}}_E^T \mathbf{K}_{P_M} \boldsymbol{\theta}_E \quad (7)$$

which after substituting the closed-loop system terms yields

$$\begin{aligned} \dot{V} = & \dot{\mathbf{q}}_f^T \left(\mathbf{C}_S \dot{\mathbf{q}}_f + \frac{\dot{\mathbf{M}}_T(\mathbf{q}) \dot{\mathbf{q}}_f}{2} + \begin{bmatrix} -\boldsymbol{\tau}_{gq} \\ 0 \end{bmatrix} - \mathbf{K}_T \mathbf{q}_f - \boldsymbol{\Psi} \dot{\mathbf{q}}_f \right) \\ & + \begin{bmatrix} 0 \\ \mathbf{K}_{P_M} \boldsymbol{\theta}_E \end{bmatrix} + \begin{bmatrix} 0 \\ \dot{\boldsymbol{\theta}}_S^T \mathbf{I} (\dot{\mathbf{q}}^T \boldsymbol{\tau}_{gq} - \dot{\boldsymbol{\theta}}_d^T \mathbf{K}_{P_M} \boldsymbol{\theta}_E) \end{bmatrix} \\ & + \dot{\mathbf{q}}_f^T \mathbf{K}_T \mathbf{q}_f - \dot{\boldsymbol{\theta}}^T \mathbf{K}_{P_M} \boldsymbol{\theta}_E + \dot{\boldsymbol{\theta}}_d^T \mathbf{K}_{P_M} \boldsymbol{\theta}_E \end{aligned} \quad (8)$$

assuming $\mathbf{C}_S = \begin{bmatrix} -\mathbf{C} & 0 \\ 0 & 0 \end{bmatrix}$, $\dot{\mathbf{q}}^T (\dot{\mathbf{M}} - 2\mathbf{C}) \dot{\mathbf{q}} = 0$ [25], [47], and $\boldsymbol{\Psi} \in \mathbb{R}^{2n \times 2n} = \text{diag}(\mathbf{N}, \mathbf{D} + \mathbf{K}_{D_M})$. Hence, one acquires

$$\begin{aligned} \dot{V} = & -\dot{\boldsymbol{\theta}}^T \mathbf{K}_{P_M} \boldsymbol{\theta}_E + \dot{\boldsymbol{\theta}}_d^T \mathbf{K}_{P_M} \boldsymbol{\theta}_E \\ & + \dot{\mathbf{q}}_f^T \left(-\boldsymbol{\Psi} \dot{\mathbf{q}}_f + \begin{bmatrix} 0 \\ \mathbf{K}_{P_M} \boldsymbol{\theta}_E \end{bmatrix} + \begin{bmatrix} 0 \\ \dot{\boldsymbol{\theta}}_S^T \mathbf{I} (-\dot{\boldsymbol{\theta}}_d^T \mathbf{K}_{P_M} \boldsymbol{\theta}_E) \end{bmatrix} \right) \end{aligned} \quad (9)$$

which may be reduced to

$$\dot{V} = \dot{\mathbf{q}}_f^T \left(-\boldsymbol{\Psi} \dot{\mathbf{q}}_f + \begin{bmatrix} 0 \\ \dot{\boldsymbol{\theta}}_S^T \mathbf{I} (-\dot{\boldsymbol{\theta}}_d^T \mathbf{K}_{P_M} \boldsymbol{\theta}_E) \end{bmatrix} \right) + \dot{\boldsymbol{\theta}}_d^T \mathbf{K}_{P_M} \boldsymbol{\theta}_E \quad (10)$$

which can be further expanded to produce the expression

$$\begin{aligned} \dot{V} = & \dot{q}_f^T \left(\frac{\dot{M}_T(q)\dot{q}_f}{2} - \begin{bmatrix} \tau_{gq} \\ 0 \end{bmatrix} - K_T q_f - \Psi \dot{q}_f + \begin{bmatrix} 0 \\ K_{P_{Mt}} \theta_E \end{bmatrix} + \right. \\ & \left. C_S \dot{q}_f + \begin{bmatrix} 0 \\ \dot{\Theta}_S I \left(\dot{q}^T \tau_{gq} - \dot{\theta}_d^T K_{P_{Mt}} \theta_E - \frac{\theta_E^T \dot{K}_{P_{Mt}} \theta_E}{2} \right) \right] \right) \\ & \dot{q}_f^T K_T q_f + \frac{\theta_E^T \dot{K}_{P_{Mt}} \theta_E}{2} - \dot{\theta}^T K_{P_{Mt}} \theta_E + \dot{\theta}_d^T K_{P_{Mt}} \theta_E. \end{aligned} \quad (26)$$

Therefore, one finally obtains

$$\dot{V}(\dot{q}_f) = -\dot{q}_f^T \Psi \dot{q}_f \quad (27)$$

as was formerly the case. GLAS could then be concluded by means of following an approach analogous to Proof 1.

C. Variable Impedance Tracking Controller Passivity

The impact of external contacts on the system's behavior is analyzed via the passivity analysis delineated next.

Theorem 3: The closed-loop system comprising formulae (1), (2), and (24), with $\tau_{ex} \neq 0$ is passive when supplied with time-varying position references (tracking control).

Proof 3: Constructing a closed-loop system by utilizing (1), (2), and (24), while considering the storage function described by (6), an array of calculations similar to those provided in the previous subsections (assuming that $\tau_{ex} \neq 0$), produces the result

$$\dot{Q} = \dot{q}^T J_e^T f_{ex} - \dot{q}_f^T \Psi \dot{q}_f \leq \dot{q}^T J_e^T f_{ex} = \dot{q}^T \tau_{ex} \quad (28)$$

where $Q = V$ and $\tau_{ex} = J_e^T f_{ex}$ with J_e^T denoting a Jacobian. Positive definiteness of Ψ is a prerequisite for validity of (28), which ensures passivity of the system [19], [27].

IV. FULL-STATE FEEDBACK VIC TRACKING CONTROLLER

A. Full-State Feedback Tracking Controller

In order to enhance robustness, improve tracking performance, and permit systematic gain generation, it is beneficial to incorporate link-side feedback into the controller design. The latter may be achieved via the full-state feedback trajectory tracking control (FSTC) scheme described as

$$\begin{aligned} \tau_m = & K_{P_M} \theta_E - K_{D_M} \dot{\theta} + K_{P_J} q_E - K_{D_J} \dot{q} \\ & + \dot{\Theta}_S I \left(-\dot{\theta}^T K_{P_J} q_E - \dot{\theta}_d^T K_{P_M} \theta_E \right. \\ & \left. + \dot{q}_f^T K_T q_{f_d} + \dot{q}_{f_d}^T K_T q_{f_e} + \dot{q}^T \tau_{gq} - \frac{\dot{q}_{f_e}^T T_L q_{f_e}}{2} \right) \end{aligned} \quad (29)$$

where $T_L = \begin{bmatrix} 0 & K_{P_J} \\ K_{P_J} & 0 \end{bmatrix}$ and $K_{P_J} \in \mathbb{R}^{n \times n}$, $K_{D_J} \in \mathbb{R}^{n \times n}$ denote symmetric link position and velocity gain matrices. For this controller, one should initially define link position trajectories

(link trajectory tracking), via the q_d variable, which are then to be converted into referential motor positions using

$$\theta_d = K^{-1} (M(q_d)\ddot{q}_d + (C(q_d, \dot{q}_d) + N) \dot{q}_d + K q_d + \tau_{gd}) \quad (30)$$

where $\tau_{gd} = \tau_g(q_d)$. The following modified Lyapunov function can then be considered for the stability analysis:

$$V = \frac{1}{2} \dot{q}_f^T M_T \dot{q}_f + \frac{1}{2} q_{f_e}^T T_F q_{f_e} \quad (31)$$

where the overall stiffness matrix possesses the structure:

$$T_F = \begin{bmatrix} K & -K \\ -K + K_{P_J} & K + K_{P_M} \end{bmatrix}. \quad (32)$$

To ensure that $V \geq 0$, T_F should be converted into a symmetric equivalent, in the following manner:

$$\frac{1}{2} \begin{bmatrix} W + W^T & (Y + X)^T \\ Y + X & Z + Z^T \end{bmatrix} > 0 \quad (33)$$

where $W = K$, $X = -K$, $Y = -K + K_{P_J}$, and $Z = K + K_{P_M}$. By subsequently calculating the Schur complement of the above matrix's Z element, the following condition arises:

$$\Gamma = W - \frac{(Y + X)^T Z^{-1} (Y + X)}{4} > 0 \quad (34)$$

leading to establishment of the inequality $\lambda_m(\Gamma) > 0$, the satisfaction of which guarantees T_F 's positive definiteness.

Theorem 4: The closed-loop system comprising formulae (1), (2), and (29) is GAS when supplied with time-varying trajectories (tracking control) while assuming that $\tau_{ex} = 0$.

Proof 4: Computing the time derivative of the Lyapunov function described by (31) yields

$$\begin{aligned} \dot{V} = & \frac{\dot{q}_f^T \dot{M}_T \dot{q}_f}{2} + \dot{q}_f^T M_T \ddot{q}_f + \dot{q}_{f_e}^T K_T q_{f_e} \\ & + \frac{\dot{q}_{f_e}^T T_L q_{f_e}}{2} + \dot{\theta}_E^T K_{P_M} \theta_E \end{aligned} \quad (35)$$

which can be further expanded to produce

$$\begin{aligned} \dot{V} = & \dot{q}_f^T \left(C_S \dot{q}_f + \frac{1}{2} \dot{M}_T(q) \dot{q}_f + \begin{bmatrix} -\tau_{gq} \\ 0 \end{bmatrix} \right. \\ & \left. - K_T q_f - \Psi_L \dot{q}_f + \begin{bmatrix} K_{P_M} \theta_E + K_{P_J} q_E \\ 0 \end{bmatrix} + \right. \\ & \left. \left[\left(\dot{\Theta}_S I \left(\dot{q}^T \tau_{gq} - \dot{\theta}_d^T K_{P_M} \theta_E + \dot{q}_f^T K_T q_{f_d} \right) \right. \right. \right. \\ & \left. \left. + \dot{q}_{f_d}^T K_T q_{f_e} - \dot{\theta}^T K_{P_J} q_E - \frac{\dot{q}_{f_e}^T T_L q_{f_e}}{2} \right) \right] \right) \\ & + \dot{q}_{f_e}^T K_T q_{f_e} + \dot{\theta}_E^T K_{P_M} \theta_E + \frac{\dot{q}_{f_e}^T T_L q_{f_e}}{2} \end{aligned} \quad (36)$$

where $\Psi_L = \begin{bmatrix} N & 0 \\ K_{D_J} & D + K_{D_M} \end{bmatrix}$. Further simplifications yield

$$\dot{V}(\dot{q}_f) = -\dot{q}_f^T \Psi_L \dot{q}_f \quad (37)$$

where Ψ_L 's positive definiteness can be guaranteed if it satisfies a condition that is analogous to (34). GLAS can then be proven by differentiating (37) to produce

$$\ddot{V}(\dot{q}_f, \ddot{q}_f) = -\ddot{q}_f^T \Psi_L \dot{q}_f - \dot{q}_f^T \Psi_L \ddot{q}_f \quad (38)$$

where $\|\Psi_L\| \leq \psi_b$, with ψ_b denoting a positive constant.

By following the steps outlined in Proof 1 while recalling that $V > 0$ and $\dot{V} \leq 0$ implies boundedness of the system's states [48], then according to the NIP the system is GAS at q_{eq} .

B. Full-State Feedback Tracking VIC Scheme

The FSTC scheme can be rendered capable of stable VIC, via the full-state feedback trajectory tracking VIC (FSTVIC) scheme, which is shown as

$$\begin{aligned} \tau_m = & \mathbf{K}_{P_{Mt}} \theta_E - \mathbf{K}_{D_{Mt}} \dot{\theta} + \mathbf{K}_{P_{Jt}} q_E - \mathbf{K}_{D_{Jt}} \dot{q} \\ & + \dot{\theta}_S I \left(-\frac{(\dot{q}_{f_e}^T \mathbf{T}_L + q_{f_e}^T \dot{\mathbf{T}}_F) q_{f_e}}{2} + \dot{q}^T \tau_{gq} \right. \\ & - \dot{\theta}_d^T \mathbf{K}_{P_{Mt}} \theta_E - \dot{\theta}^T \mathbf{K}_{P_{Jt}} q_E + \dot{q}_f^T \mathbf{K}_T q_{f_d} \\ & \left. + \dot{q}_{f_d}^T \mathbf{K}_T q_{f_e} \right) \end{aligned} \quad (39)$$

where $\mathbf{K}_{P_{Jt}} \in \mathbb{R}^{n \times n}$ and $\mathbf{K}_{D_{Jt}} \in \mathbb{R}^{n \times n}$ denote symmetric matrices of time-varying link position and velocity gains, which can be regulated in real-time by the user.

Theorem 5: The closed-loop system comprising formulae (1), (2), and (39) is GAS when supplied with time-varying trajectories (tracking control) while assuming that $\tau_{ex} = 0$.

Proof 5: Computing the time derivative of (31), along the closed-loop trajectories yields the relationship

$$\begin{aligned} \dot{V} = & \dot{q}_f^T \left(\begin{aligned} & C_S \dot{q}_f + \frac{1}{2} \dot{\mathbf{M}}_T(q) \dot{q}_f + \begin{bmatrix} -\tau_{gq} \\ 0 \end{bmatrix} \\ & - \mathbf{K}_T q_f - \Psi_L \dot{q}_f + \begin{bmatrix} 0 \\ \mathbf{K}_{P_{Mt}} \theta_E + \mathbf{K}_{P_{Jt}} q_E \end{bmatrix} + \\ & \begin{bmatrix} \left(\dot{\theta}_S I \left(\dot{q}^T \tau_{gq} - \dot{\theta}_d^T \mathbf{K}_{P_{Mt}} \theta_E \right. \right. \\ \left. \left. - \frac{q_{f_e}^T \dot{\mathbf{T}}_F q_{f_e}}{2} - \dot{\theta}^T \mathbf{K}_{P_{Jt}} q_E \right. \right. \\ \left. \left. + \dot{q}_f^T \mathbf{K}_T q_{f_d} + \dot{q}_{f_d}^T \mathbf{K}_T q_{f_e} - \frac{\dot{q}_{f_e}^T \mathbf{T}_L q_{f_e}}{2} \right) \right] \end{aligned} \right) \\ & + \dot{q}_{f_e}^T \mathbf{K}_T q_{f_e} + \dot{\theta}_E^T \mathbf{K}_{P_{Mt}} \theta_E + \frac{\dot{q}_{f_e}^T \mathbf{T}_L q_{f_e}}{2} + \frac{q_{f_e}^T \dot{\mathbf{T}}_F q_{f_e}}{2} \end{aligned} \quad (40)$$

where $\Psi_F = \begin{bmatrix} \mathbf{N} & 0 \\ \mathbf{K}_{D_{Jt}} \mathbf{D} + \mathbf{K}_{D_{Mt}} \end{bmatrix}$. An array of simplifications finally yields the following equation:

$$\dot{V}(\dot{q}_f) = -\dot{q}_f^T \Psi_F \dot{q}_f \quad (41)$$

as was formerly the case. The gain selection process should ensure positive definiteness of Ψ_F . GLAS is concluded by following an approach analogous to that outlined in Proof 1.

C. Full-State Feedback VIC Passivity

Since this controller aims at rendering robotic devices safe for human-robot (or environmental) interaction, it is crucial to conduct a passivity analysis to corroborate the latter feature.

Theorem 6: The closed-loop system comprising formulae (1), (2), and (39), with $\tau_{ex} \neq 0$ is passive when supplied with time-varying position references (tracking control).

Proof 6: Considering the closed-loop system comprising (1), (2), and (39) together with storage function (31), an array of calculations similar to those provided in the previous subsections (assuming that $\tau_{ex} \neq 0$), produces

$$\dot{Q} = \dot{q}^T \tau_{ex} - \dot{q}_f^T \Psi_F \dot{q}_f \leq \dot{q}^T \tau_{ex}. \quad (42)$$

The positive definiteness of Ψ_F is a prerequisite for validity of (42), and can be guaranteed by appropriate gain selection.

D. Systematic Full-State Feedback Gain Generation

Linearization about a desired operating point leads to production of the following equations:

$$\mathbf{M}_L \ddot{q} + \mathbf{N}_L \dot{q} + \mathbf{K}(q - \theta) + \tau_{g_L} = 0 \quad (43)$$

$$\mathbf{J} \ddot{\theta} + \mathbf{D} \dot{\theta} + \mathbf{K}(\theta - q) = \tau_m \quad (44)$$

where $\mathbf{M}_L, \mathbf{N}_L \in \mathbb{R}^{n \times n}$, and $\tau_{g_L} \in \mathbb{R}^n$ denote invariant terms resulting from linearization of $\mathbf{M}(q)$, $\mathbf{N}$, and τ_{gq} . Thus, a state-space representation may be provided as

$$\dot{x} = \begin{bmatrix} 0 & \mathbf{I} \\ -\mathbf{M}_{L_T}^{-1} \mathbf{K}_L & -\mathbf{M}_{L_T}^{-1} \mathbf{N}_{L_T} \end{bmatrix} x + \begin{bmatrix} 0 \\ \mathbf{M}_{L_T}^{-1} \mathbf{E}_s \end{bmatrix} \tau_m \quad (45)$$

where $\mathbf{M}_{L_T} = \text{diag}(\mathbf{M}_L, \mathbf{J})$, $\mathbf{N}_{L_T} = \text{diag}(\mathbf{N}_L, \mathbf{D})$, and $\mathbf{K}_L$ denotes the linearized potential matrix, containing the passive elasticity and gravitational terms. Moreover, $\mathbf{E}_s \in \mathbb{R}^{2n \times n}$ symbolizes a selection matrix and $x = [q^T \theta^T \dot{q}^T \dot{\theta}^T]^T$, thus resulting in the following state-space form:

$$\dot{x} = \mathbf{A}x + \mathbf{B}u \quad (46)$$

$$y = \mathbf{C}x \quad (47)$$

where $\mathbf{C} = [\mathbf{I} \ 0]$, $u = \tau_m$, and $\mathbf{A} \in \mathbb{R}^{4n \times 4n}$, $\mathbf{B} \in \mathbb{R}^{4n \times n}$ denote the state transition and input matrices. The centralized LQR controller design entails minimization of the following cost function subject to the system's dynamics:

$$S = \sum_{k=0}^{\infty} \{x^T \mathbf{Q}x + \tau_m^T \mathbf{R} \tau_m\} \quad (48)$$

where $\mathbf{Q} \in \mathbb{R}^{4n \times 4n}$ and $\mathbf{R} \in \mathbb{R}^{n \times n}$ denote diagonal and positive definite penalty matrices. This optimization yields a solution to the following algebraic Riccati equation:

$$\mathbf{A}^T \mathbf{P}_u + \mathbf{P}_u \mathbf{A} - \mathbf{P}_u \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T \mathbf{P}_u + \mathbf{Q} = 0 \quad (49)$$

where $\mathbf{P}_u \in \mathbb{R}^{4n \times 4n}$ denotes the unknown matrix, whose value is utilized to compute the following gain matrix:

$$\mathbf{K}_{LQR} = \mathbf{R}^{-1} \mathbf{B}^T \mathbf{P}_u \quad (50)$$

where $\mathbf{K}_{LQR} = [\mathbf{K}_{P_J} \ \mathbf{K}_{P_M} \ \mathbf{K}_{D_J} \ \mathbf{K}_{D_M}]$. The generated gains are then decentralized [49], [50] and appended to the control signals to yield (39). Satisfaction of $\lambda_m(\Gamma) > 0$ and $\Psi_F \geq 0$ can generally be attained by assigning larger values to the $\mathbf{Q}$ matrix elements (penalties) associated with motor terms.



Fig. 2. Snapshots of a human operator interacting with a Baxter robot.

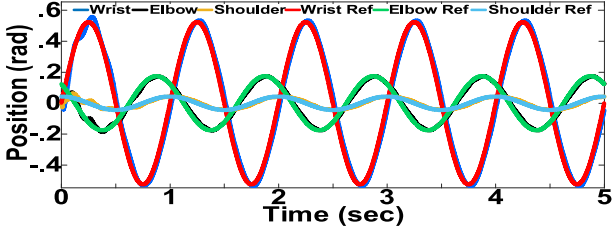


Fig. 3. Simulated link tracking (1 Hz input, constant impedance).

V. VARIABLE IMPEDANCE AND TRACKING EXPERIMENTS

The scheme's efficacy has been corroborated by means of simulations revolving around a sagittal, flexible-joint robotic arm model. Moreover, experimental results have been extracted from a collaborative robot, namely the Rethink Robotics Baxter (Fig. 2), which is powered by compliant SEAs and controlled via the FSTVIC scheme. Baxter's shoulder/elbow, and wrist joints possess flexure stiffness values of 843, and 250 Nm/rad, respectively [51]. The experimental setup entails an Intel Core i7-6820HQ @ 2.7 GHz desktop PC, running Robot Operating System (ROS) Kinetic, through Ubuntu 14.04 LTS (Trusty Tahr).

A. Compliant Robot Trajectory-Tracking Simulation

Simulation of a sagittal, 3-DoF, flexible-joint robot has been performed by means of numerically integrating (1) and (2), within a loop executed at a sampling rate corresponding to that utilized for the real experiments (~ 1 ms). Baxter's dynamical parameter values have been obtained from two sources [51], [52]. The simulated robot is supplied with a desired sinusoidal joint-position trajectory generated in accordance with the following equation:

$$q_{d_i} = q_i - A_{m_i} \cdot \sin(2\pi f t + \pi_{o_i}) \quad (51)$$

where i , A_m , f , t , and π_o signify the i th joint, amplitude, frequency, time, and phase shift offset, respectively.

Equation (51) permits defining distinct oscillatory motions for each joint, characterized by different amplitudes and phase offsets. Fig. 3 displays the joint tracking performance corresponding to the three pitch joints of the flexible-joint, robotic arm model underpinning the simulation. The following LQR

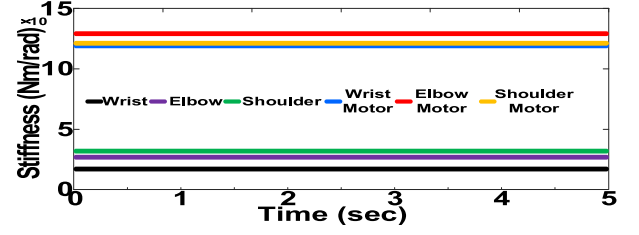


Fig. 4. Constant stiffness gain values.

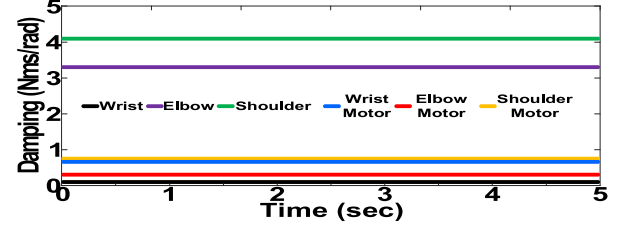


Fig. 5. Constant damping gain values.



Fig. 6. Baxter performing sinusoidal motions during tracking experiments.

penalties

$$\mathbf{Q}_H = \text{diag} \left(\begin{bmatrix} 1e^3 & 1.875 & 1.73 & 1.781 & 8.72 & 7e^{-5} & 1e^{-6} & \dots \\ \dots & 0.04 & 1 & 19 & 0.8 & 15 & 6 & 4 & 0.4 & \mathbf{I}_{1 \times 14} \cdot e^{-18} \end{bmatrix} \right)$$

$$\mathbf{R}_H = \text{diag}([111110.25 \ 1])$$

led to production of the high-impedance $\mathbf{K}_{P_J}$ (wrist, elbow, shoulder), $\mathbf{K}_{P_M}$ (wrist motor, elbow motor, shoulder motor), $\mathbf{K}_{D_J}$, and $\mathbf{K}_{D_M}$ gains depicted in Figs. 4 and 5, respectively.

B. Compliant Robot Trajectory-Tracking Experiments

Prior to executing the trajectory-tracking task, Baxter's arms are initially resting at predefined home configurations (Fig. 6, leftmost picture) while bearing no additional loads.

Subsequently, desired sinusoidal motions are excited via exploitation of (51). The link position feedback variable, $\mathbf{q}$, required for computation of $\dot{\mathbf{q}}$, and by extension (51), is calculated as

$$\mathbf{q} = \mathbf{K}^{-1} \boldsymbol{\tau} + \boldsymbol{\theta}. \quad (52)$$

$\boldsymbol{\tau} \in \mathbb{R}^n$ denotes the vector of deflection torques, which can be acquired from the torque sensors embedded in Baxter's joints. Hence, the link velocity variables may be acquired via numerical differentiation as demonstrated below

$$\dot{\mathbf{q}}_j = \frac{\mathbf{q}_j - \mathbf{q}_{j-1}}{\Delta t} \quad (53)$$

where Δt and j denote the sampling interval and j th sample.

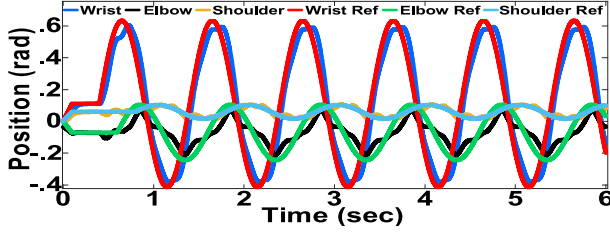


Fig. 7. Right arm link tracking (1 Hz input signal, constant impedance).

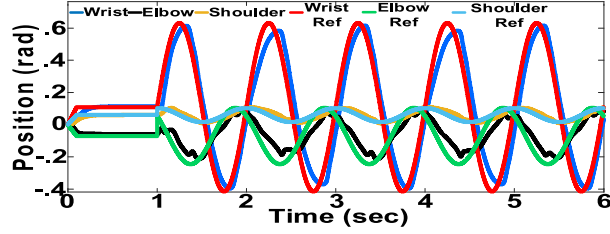


Fig. 8. Right arm link tracking (1 Hz input signal, variable impedance).

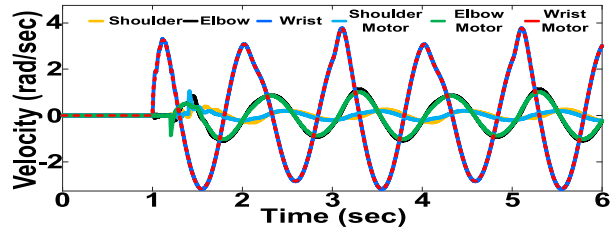


Fig. 9. Right arm link and motor velocities.

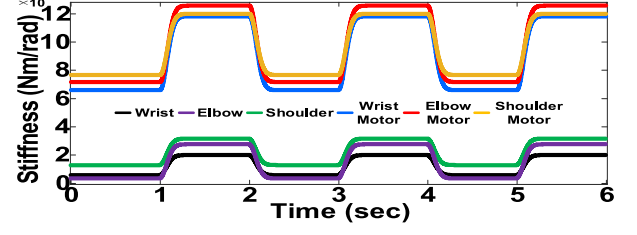


Fig. 10. Active stiffness gain modulation.

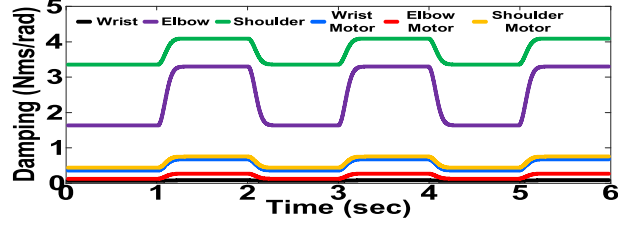


Fig. 11. Active damping gain modulation.

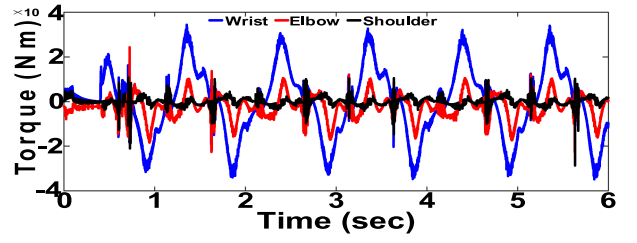


Fig. 12. Control torque signals produced during trajectory execution.

The Baxter was initially provided with sinusoidal position inputs while possessing constant impedance gains identical to those depicted in Figs. 4 and 5. Fig. 7 exhibits the desired and actual sagittal link position variables (q), extracted during the robot's tracking of a predefined position trajectory which was executed while the robot's impedance gains (K_{PJ} , K_{PM} , K_{DJ} , K_{DM}) remained constant. Fig. 8 displays the desired and actual, link position variables measured during trajectory tracking, which was performed while the robot's impedance gains were varied in predetermined fashion, bereft of any feedback-based switching laws. This revealed the FSTVIC method's efficacy in terms of maintaining stability during VIC. Fig. 9 presents the link velocities produced during the VIC tracking experiment pertaining to Fig. 8. The lower impedance gain limits, depicted in Figs. 10 and 11, were generated using the following LQR penalties:

$$\mathbf{Q}_L = \text{diag} \left(\begin{matrix} 1e^3 & [1.875 & 0.13 & 0.181 & 0.02 & 7e^{-5} & 1e^{-6} & \dots] \\ \dots & 0.04 & 1 & 18 & 1.8 & 12 & 6 & 2 & 0.4 & \mathbf{I}_{1 \times 14} \cdot e^{-18} \end{matrix} \right)$$

$$\mathbf{R}_L = \text{diag} (2 \cdot [1 \ 1 \ 1 \ 1 \ 1 \ 0.25 \ 1]).$$

Fig. 12 illustrates the input torques generated during the variable impedance tracking experiments. The control loop's sampling time values fluctuated between 1 and 3 ms.

C. Compliant Robot Variable Impedance Control

A ramp function is exploited to render the robot compliant once an external force has been detected, as described below:

$$\mathbf{K}_{LQR_k} = \begin{cases} \mathbf{K}_{LQR_{k-1}} + (\mathbf{K}_H - \mathbf{K}_{LQR_{k-1}})k/t_r & \text{if } \tau_{ex} = 0 \\ \mathbf{K}_{LQR_{k-1}} - (\mathbf{K}_{LQR_{k-1}} - \mathbf{K}_L)k/t_r & \text{if } \tau_{ex} \neq 0 \end{cases} \quad (54)$$

where k symbolizes the k th sample, t_r signifies the ramping duration, while $\mathbf{K}_H$, $\mathbf{K}_L \in \mathbb{R}^{2n \times 2n}$ denote high and low impedance gain sets. These gains are only optimal at the linearization points, which is acceptable, since FSTVIC does not impose gain constraints. For demonstrative purposes, the left arm serves as an impedance regulation trigger for the right hand (Fig. 2). Thus, exceedance of a particular left-hand external force threshold induces a reduction in the right arm's active impedance values. Analogously, the right arm's active impedances are gradually increased, in the absence of external forces, or when the external forces possess values lower than the defined thresholds. Fig. 13–16 exhibit the gain values pertaining to the right arm during the interaction experiment. To be precise, Figs. 13–16 display the active motor stiffness (K_{PM}), link stiffness (K_{PJ}), motor damping (K_{DM}), and link damping gain (K_{DJ}) modulation, respectively. These gain transitions were dictated by the left arm's contact state, which served as an “impedance trigger.” Fig. 17 reveals that the values contained in

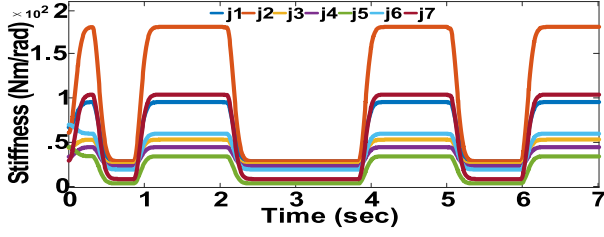


Fig. 13. Active motor stiffness modulation during interaction (right arm).

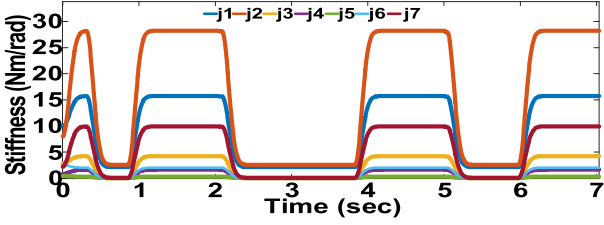


Fig. 14. Active link stiffness modulation during interaction (right arm).

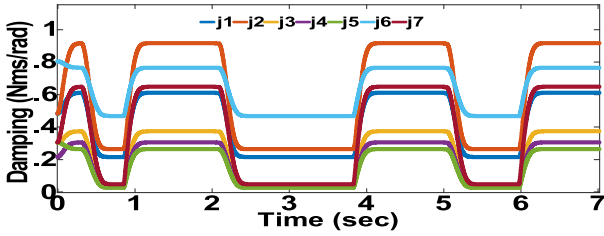


Fig. 15. Active motor damping modulation during interaction (right arm).

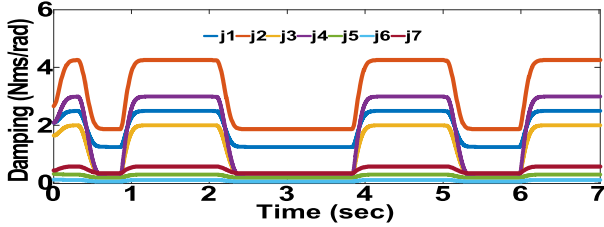


Fig. 16. Active link damping modulation during interaction (right arm).

Figs. 13–16 decrease, as the values of Fig. 17 increase (contact), while the former values increase as the wrenches in Fig. 17 decrease (no contact). Fig. 18 exhibits the impact forces applied by the user onto the right arm to physically gauge the robot's impedance levels. Fig. 19 contains the input torques produced by (39) during interaction, while Fig. 20 displays the link velocity values measured during interaction.

D. Controller Feasibility Discussion

τ_m is practically bounded since the range of $\dot{\theta}_S$ is closed, i.e., limited by the encoder resolution. Thus, $\dot{\theta}_{S_{\max}} = \frac{1}{v_m}$, where v_m

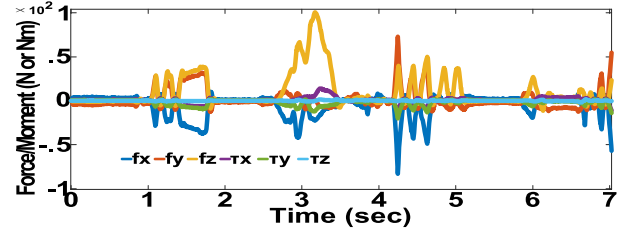


Fig. 17. External wrenches produced during interaction (left arm).

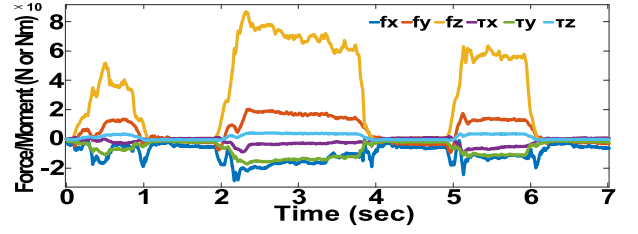


Fig. 18. External wrenches produced during interaction (right arm).

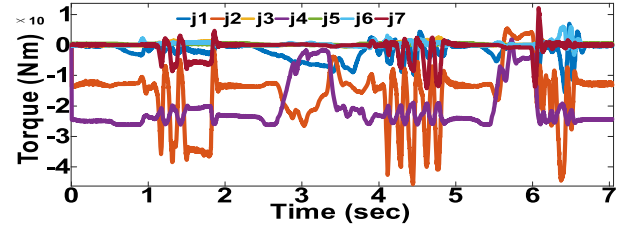


Fig. 19. Control input torques produced during interaction (right arm).

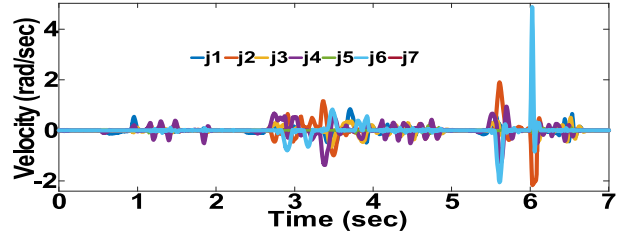


Fig. 20. Joint velocity evolution during interaction (right arm).

denotes the minimum velocity measurement. $\max(\tau_m) \geq \tau_{\max}$ and $\min(\tau_m) \leq -\tau_{\max}$ can always be prevented if they occur when $a = \dot{q}^T \tau_{gq} - \dot{\theta}_d^T K_{PM} \theta_E > 0$, since one may define $s_P = \dot{\theta}_S I(a - c_s)$ in controller (5), with $c_s = q_f^T K_G(t) q_{fe}$ and $K_G(t) > 0$, thereby ensuring $\dot{V} = -\dot{q}_f^T \Psi \dot{q}_f - c_s$. If $a < 0$, saturation prevention requires stiffness modulations, such that $s_P = \dot{\theta}_S I(a - c_s)$, with $c_s = \frac{\theta_E^T K_{PM} \theta_E}{2}$ and $\dot{K}_{PM} < 0$, thus yielding $\dot{V} = -\dot{q}_f^T \Psi \dot{q}_f$. Assuming identical stiffness fluctuations for all K_{PM} gains, the required gain-modulation rate can be computed as follows:

$$a + m_r \frac{\theta_E^T I_d \theta_E}{2} = c_{th} \Rightarrow m_r = \frac{2(c_{th} - a)}{\theta_E^T I_d \theta_E} \quad (55)$$

where $\mathbf{I}_d$ is an identity matrix, and c_{th} is a user-selected variable that when multiplied by $\dot{\Theta}_S \mathbf{I}$ yields τ_{mf} , a feasible input signal. Practically, $\dot{\mathbf{K}}_{P_M}$'s values can be large, since $\dot{\mathbf{K}}_{P_{M_i}} = \frac{1}{\Delta t} (\mathbf{K}_{P_{M_i}} - \mathbf{K}_{P_{M_{i-1}}})$. Hence, $\frac{\theta_E^T \dot{\mathbf{K}}_{P_{M_i}} \theta_E}{2} \geq a$ should always be achievable. It is noteworthy that discontinuities have no effect on closed-loop stability owing to the positive definiteness of V , and the negative semidefiniteness of $\dot{V}$. The latter conclusions conform to Theorem 2.3 in [48], and the relevant statements on page 193 of [48].

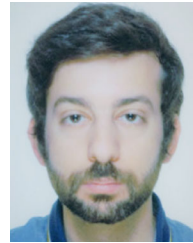
VI. CONCLUSION

This article provided a theoretically sound solution to the quandary of designing a flexible-joint robot control scheme that is capable of stable trajectory tracking during free motion, and passivity preservation during performance of VIC. The MMTVIC approach provided an analytical solution to the latter issue without imposing explicit restrictions on the active gains' magnitudes. Moreover, in contrast to the preponderance of flexible-joint robot control schemes, the proposed motor-side controller (MMTVIC) offers an almost model-free solution, as it relies solely upon knowledge of the gravitational torque vector. The stability and passivity analyses provided herein are numerically and experimentally validated via an array of trajectory tracking and VIC-based interaction experiments involving the Baxter.

REFERENCES

- [1] B. Siciliano and O. Khatib, *Springer Handbook of Robotics*, 2nd ed., Berlin, Germany: Springer-Verlag, 2016.
- [2] N. Hogan, "Impedance control: An approach to manipulation," in *Proc. Amer. Control Conf.*, 1984, pp. 304–313.
- [3] N. Hogan, "On the stability of manipulators performing contact tasks," *IEEE J. Robot. Autom.*, vol. 4, no. 6, pp. 677–686, Dec. 1988.
- [4] A. Dietrich, X. Wu, K. Bussmann, C. Ott, A. Albu-Schäffer, and S. Stramigioli, "Passive hierarchical impedance control via energy tanks," *IEEE Robot. Autom. Lett.*, vol. 2, no. 2, pp. 522–529, Apr. 2017.
- [5] D. Lee and K. Huang, "Passive-set-position-modulation framework for interactive robotic systems," *IEEE Trans. Robot.*, vol. 26, no. 2, pp. 354–369, Apr. 2010.
- [6] J. Buchli, F. Stulp, E. Theodorou, and S. Schaal, "Learning variable impedance control," *Int. J. Robot. Res.*, vol. 30, no. 7, pp. 820–833, 2011.
- [7] R. Ikeura and H. Inooka, "Variable impedance control of a robot for cooperation with a human," in *Proc. IEEE Int. Conf. Robot. Autom.*, 1995, pp. 3097–3102.
- [8] T. Tsumugiwa, R. Yokogawa, and K. Hara, "Variable impedance control based on estimation of human arm stiffness for human-robot cooperative calligraphic task," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2002, pp. 644–650.
- [9] C. Yang, G. Ganesh, S. Haddadin, S. Parusel, A. Albu-Schäffer, and E. Burdet, "Human-like adaptation of force and impedance in stable and unstable interactions," *IEEE Trans. Robot.*, vol. 27, no. 5, pp. 918–930, Oct. 2011.
- [10] S. Calinon, D. Bruno, and D. Caldwell, "A task-parameterized probabilistic model with minimal intervention control," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2014, pp. 3339–3344.
- [11] Z. Li, J. Liu, Z. Huang, Y. Peng, H. Pu, and L. Ding, "Adaptive impedance control of human-robot cooperation using reinforcement learning," *IEEE Trans. Ind. Electron.*, vol. 64, no. 10, pp. 8013–8022, Oct. 2017.
- [12] M. Matinfar and K. Hashtrudi-Zaad, "Optimization-based robot compliance control: geometric and linear quadratic approaches," *Int. J. Robot. Res.*, vol. 24, no. 8, pp. 645–656, 2005.
- [13] E. Gribovskaya, A. Kheddar, and A. Billard, "Motion learning and adaptive impedance for robot control during physical interaction with humans," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2011, pp. 4326–4332.
- [14] M. Grebenstein, M. Chalon, W. Friedl, S. Haddadin, T. Wimböck, and R. Siegwart, "The hand of the DLR hand arm system: Designed for interaction," *Int. J. Robot. Res.*, vol. 31, no. 13, pp. 1531–1555, 2012.
- [15] P. Qi, C. Qiu, H. Liu, J. S. Dai, L. D. Seneviratne, and K. Althoefer, "A novel continuum manipulator design using serially connected double-layer planar springs," *IEEE/ASME Trans. Mechatronics*, vol. 21, no. 3, pp. 1281–1292, Jun. 2016.
- [16] D. J. Braun *et al.*, "Robots driven by compliant actuators: Optimal control under actuation constraints," *IEEE Trans. Robot.*, vol. 29, no. 5, pp. 1085–1101, Oct. 2013.
- [17] G. Pratt and M. Williamson, "Series elastic actuators," in *Proc. IEEE/RJ Int. Conf. Intell. Robots Syst.*, 1995, pp. 399–406.
- [18] A. Jafari N. G. Tsagarakis, I. Sardellitti, and D. G. Caldwell, "A new actuator with adjustable stiffness based on a variable ratio lever mechanism," *IEEE/ASME Trans. Mechatronics*, vol. 19, no. 1, pp. 55–63, Feb. 2014.
- [19] C. Ott, A. Albu-Schäffer, A. Kugi, and G. Hirzinger, "On the passivity-based impedance control of flexible joint robots," *IEEE Trans. Robot.*, vol. 24, no. 2, pp. 416–429, Apr. 2008.
- [20] L. Lanari, P. Sicard, and J. T. Wen, "A passivity/Lyapunov approach for the control of flexible joint manipulators," in *Proc. Amer. Control Conf.*, 1993, pp. 2887–2891.
- [21] A. Albu-Schäffer, C. Ott, and G. Hirzinger, "A passivity based Cartesian impedance controller for flexible joint robots - part ii: full state feedback, impedance design and experiments," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2004, pp. 2666–2672.
- [22] S. Nicosia and P. Tomei, "Design of global tracking controllers for flexible-joint robots," *J. Robot. Syst.*, vol. 109, pp. 310–319, 1993.
- [23] L. Tian and A. Goldenberg, "Robust adaptive control of flexible joint robots with joint torque feedback," in *Proc. IEEE Int. Conf. Robot. Autom.*, 1995, pp. 1229–1234.
- [24] F. Petit, A. Dietrich, and A. Albu-Schäffer, "Generalizing torque control concepts: using well-established torque control methods on variable stiffness robots," *IEEE Robot. Autom. Mag.*, vol. 22, no. 4, pp. 37–51, Dec. 2015.
- [25] P. Tomei, "A Simple PD controller for robots with elastic joints," *IEEE Trans. Automat. Control*, vol. 36, pp. 1208–1213, Oct. 1991.
- [26] A. de Luca, B. Siciliano, and L. Zollo, "PD control with on-line gravity compensation for robots with elastic joints: Theory and experiments," *Automatica*, vol. 41, no. 10, pp. 1809–1819, 2005.
- [27] A. van der Schaft, *L2-Gain and Passivity Theorem Techniques in Nonlinear Control*. Berlin, Germany: Springer-Verlag, no. 2, 2000.
- [28] M. Spong, "Modeling and control of elastic joint robots," *ASME J. Dyn. Syst., Measure., Control*, vol. 109, no. 4, pp. 310–318, 1987.
- [29] M. Keppeler, D. Lakatos, and C. Ott, "Elastic structure preserving (ESP) control of compliantly actuated robots," *IEEE Trans. Robot.*, vol. 34, no. 2, pp. 317–335, Apr. 2018.
- [30] A. Giusti, J. Malzahn, N. G. Tsagarakis, and M. Althoff, "Combined inverse-dynamics/passivity-based control for robots with elastic joints," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2017, pp. 5281–5288.
- [31] A. Loria and R. Ortega, "On tracking control of rigid and flexible joint robots," *Appl. Math. Comput. Sci.*, vol. 5, no. 2, pp. 101–113, 1995.
- [32] G. Palli, C. Melchiorri, and A. De Luca, "On the Feedback Linearization of Robots with Variable Joint Stiffness," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2008, pp. 1753–1759.
- [33] E. Spyros-Papastavridis, N. Kashiri, P. R. N. Childs, and N. G. Tsagarakis, "Online impedance regulation techniques for compliant humanoid balancing," *Robot. Autom. Syst.*, vol. 104, pp. 85–98, 2018.
- [34] E. Spyros-Papastavridis, N. Kashiri, J. Lee, N. G. Tsagarakis, and D. G. Caldwell, "Online impedance parameter tuning for compliant biped balancing," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2015, pp. 210–216.
- [35] E. Spyros-Papastavridis, P. R. N. Childs, and N. G. Tsagarakis, "Variable impedance walking using time-varying Lyapunov stability margins," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2017, pp. 318–323.
- [36] F. Wang, H. An, H. Ma, C. Li, and Q. Wei, "Active variable impedance control for hydraulic-driving articulated robotic leg," in *Proc. 13th World Congr. Intell. Control Autom.*, 2018, pp. 1561–1565.

- [37] C. Semini *et al.*, "Towards versatile legged robots through active impedance control," *Int. J. Robot. Res.*, vol. 34, no. 7, pp. 1003–1020, 2015.
- [38] S. Hussein, S. Q. Xie, and P. K. Jamwal, "Adaptive impedance control of a robotic orthosis for gait rehabilitation," *IEEE Trans. Cybern.*, vol. 43, no. 3, pp. 1025–1034, Jun. 2013.
- [39] H. F. N. Al-Shuka, S. Leonhardt, W. -H. Zhu, C. Ding, and Y. Li, "Active impedance control of bioinspired motion robotics manipulators: An overview," *Appl. Bionics Biomechanics*, 2018.
- [40] K. Kronander and A. Billard, "Stability considerations of variable impedance control," *IEEE Trans. Robot.*, vol. 32, no. 5, pp. 1298–1305, Oct. 2016.
- [41] F. Ferraguti, C. Secchi, and C. Fantuzzi, "A tank-based approach to impedance control with variable stiffness," in *Proc. IEEE Int. Conf. Robot. Autom.*, pp. 4948–4953, 2013.
- [42] F. Ferraguti *et al.*, "An energy tank-based interactive control architecture for autonomous and teleoperated robotic surgery," *IEEE Trans. Robot.*, vol. 31, no. 5, pp. 1073–1088, Oct. 2015.
- [43] J. S. Shamma and M. Athans, "Gain scheduling: Potential hazards and possible remedies," *IEEE Control Syst. Mag.*, vol. 12, no. 3, pp. 101–107, Jun. 1992.
- [44] E. Spyrakos-Papastavridis, P. R. N. Childs, and J. S. Dai, "Passivity preservation for variable impedance control of compliant robots," *IEEE/ASME Trans. Mechatronics*, to be published, doi: 10.1109/TMECH.2019.2961478.
- [45] R. Tedrake, "Underactuated robotics: Algorithms for walking, running, swimming, flying, and manipulation (course notes for MIT 6.832)," [Online]. Available: <http://underactuated.mit.edu/>.
- [46] S. Rudra, R. K. Barai, and M. Maitra, "Nonlinear state feedback controller design for underactuated mechanical system: A modified block backstepping approach," *ISA Trans.*, vol. 53, pp. 317–326, 2014.
- [47] J. E. Slotine and W. Li, "On the adaptive control of robot manipulators," *Int. J. Robot. Res.*, vol. 6, pp. 49–59, 1987.
- [48] I. Barkana, "Defending the beauty of the invariance principle," *Int. J. Control*, vol. 87, no. 1, pp. 186–206, 2014.
- [49] E. Spyrakos-Papastavridis, J. S. Dai, P. R. N. Childs, and N. G. Tsagarakis, "Selective-compliance-based Lagrange model and multilevel noncollocated feedback control of a humanoid robot," *J. Mechanisms Robot.*, vol. 10, no. 3, 2018, Art. no. 031009.
- [50] E. Spyrakos-Papastavridis, D. G. Caldwell, and N. G. Tsagarakis, "Balance and impedance optimization control for compliant humanoid stepping," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2016, pp. 1349–1355.
- [51] [Online]. Available: https://sdk.rethinkrobotics.com/wiki/Hardware_Specifications#Joint_Flexure_Stiffness
- [52] M. Bagheri and P. Naseradinmousavi, "Novel analytical and experimental trajectory optimization of a 7-DOF Baxter robot: Global design sensitivity and step size analyses," *Int. J. Adv. Manuf. Technol.*, vol. 93, no. 9–12, pp. 4153–4167, 2017.



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